Sales Performance & Forecasting Analysis

Python & Data Science Internship Assessment

Project Objective

This project performs an end-to-end analysis of the Superstore retail sales dataset. The analysis covers data cleaning, exploratory data analysis, statistical analysis, and machine-learning-based sales forecasting.

Technologies Used

Python 3, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn and Jupyter Notebook.

Dataset

Superstore Sales Dataset — Kaggle.

1. Project Description

In this project, I analyzed the Superstore sales data to understand sales patterns and tried to predict monthly sales using machine learning. The workflow includes inspection and cleaning of the raw dataset, exploratory visualizations, statistical summaries, predictive modeling, model evaluation, and business recommendations.

2. Data Cleaning & Preparation

The dataset contains 9,800 rows and 18 columns. The data was inspected for missing values, duplicate records, and incorrect data types.

There were 11 missing values in the Postal Code column. These values were replaced with the label 'Unknown' because Postal Code is a location identifier and replacing it with a numerical average would not be meaningful.

The Order Date and Ship Date columns were converted from text to datetime format. After conversion, both date columns contained zero missing values. No duplicate rows were found in the dataset.

3. Statistical Summary

Statistic	Sales
Mean	230.77
Median	54.49
Standard Deviation	626.65
Minimum	0.444
Maximum	22,638.48

Interpretation: The mean sales value of 230.77 is substantially higher than the median of 54.49. This indicates that a relatively small number of high-value orders strongly influence the average sales value.

4. Code and Output Screenshots

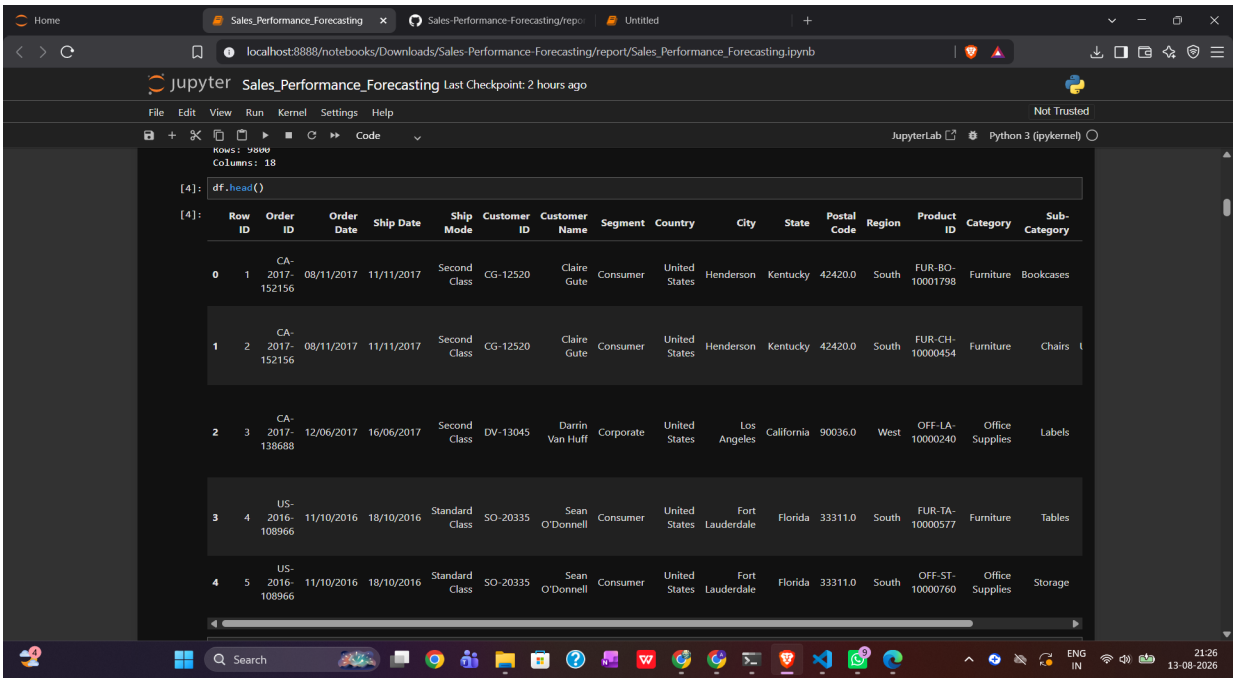

Figure 1 — Dataset Preview

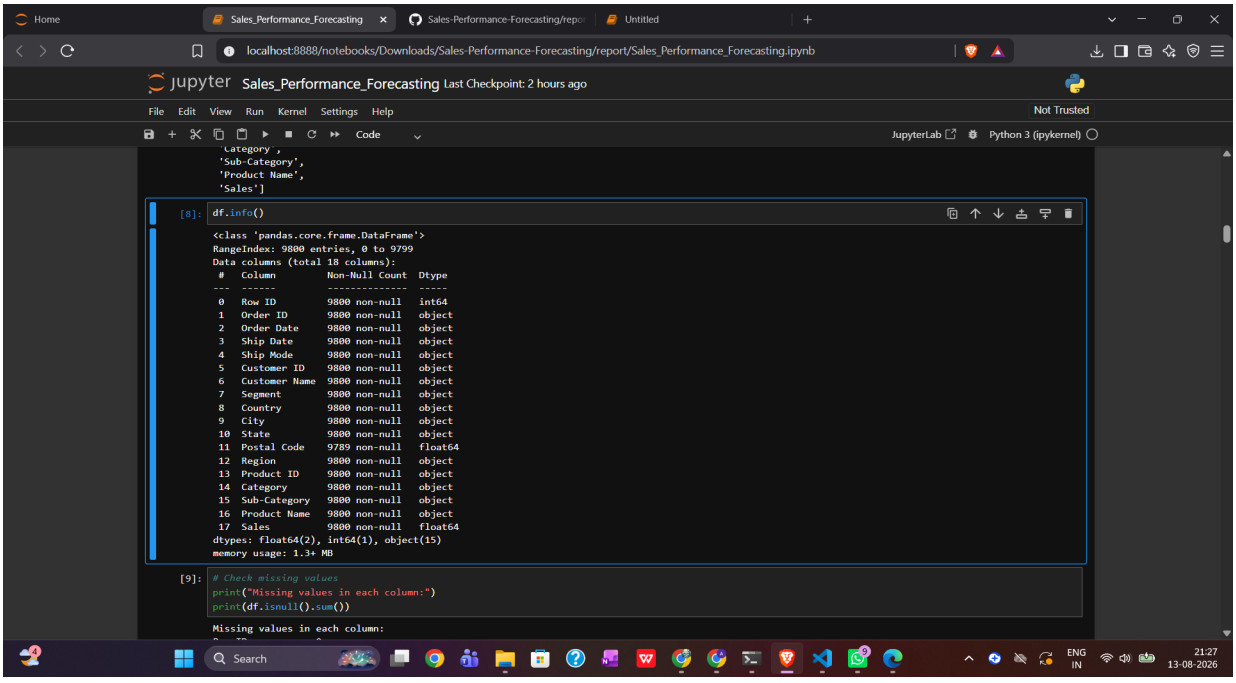

Figure 2 — Dataset Information

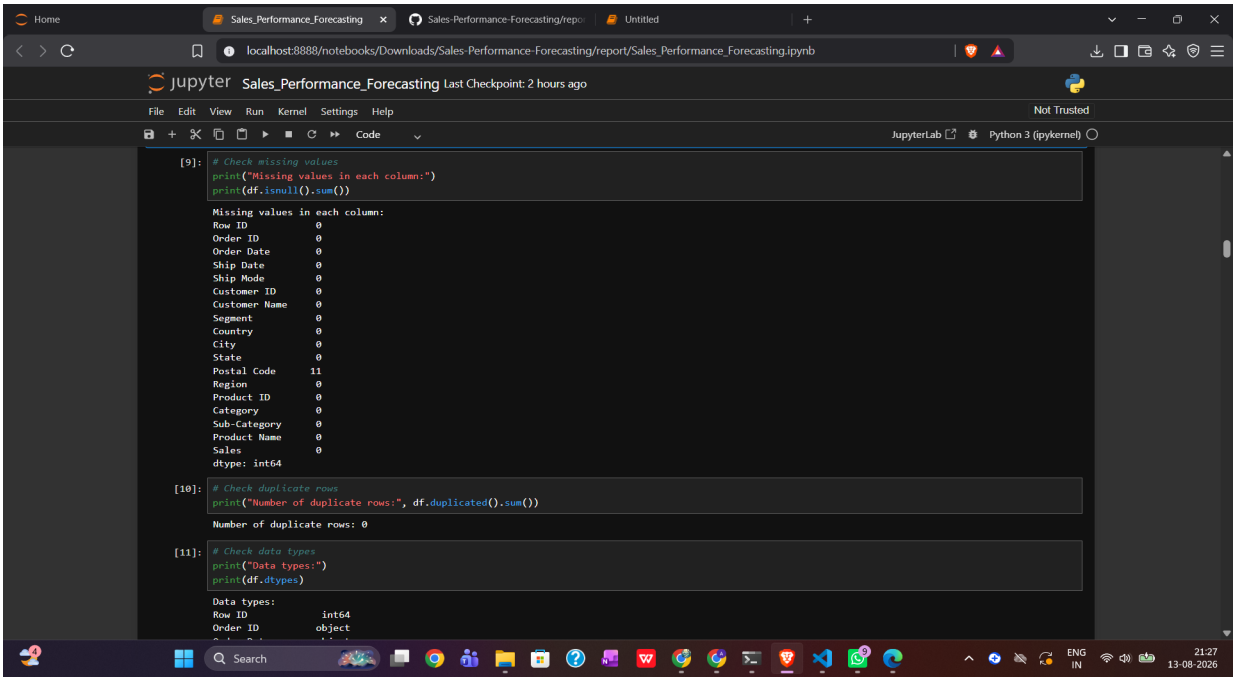

Figure 3 — Missing Value Check

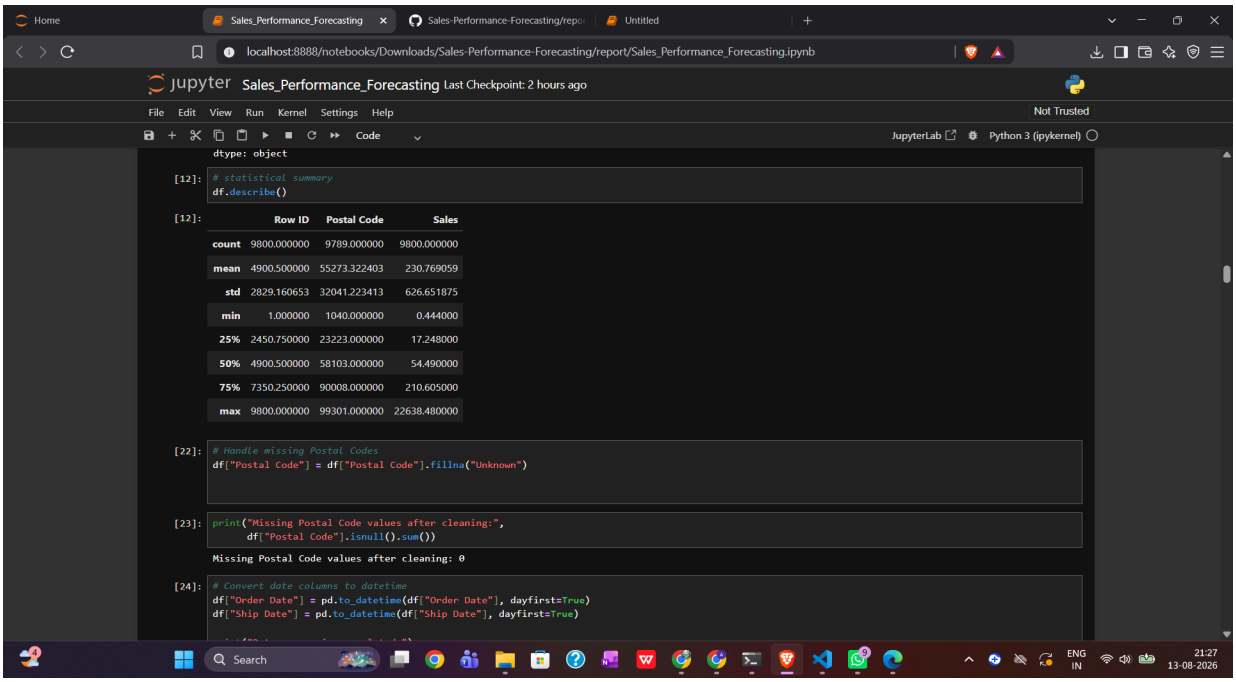

Figure 4 — Statistical Summary

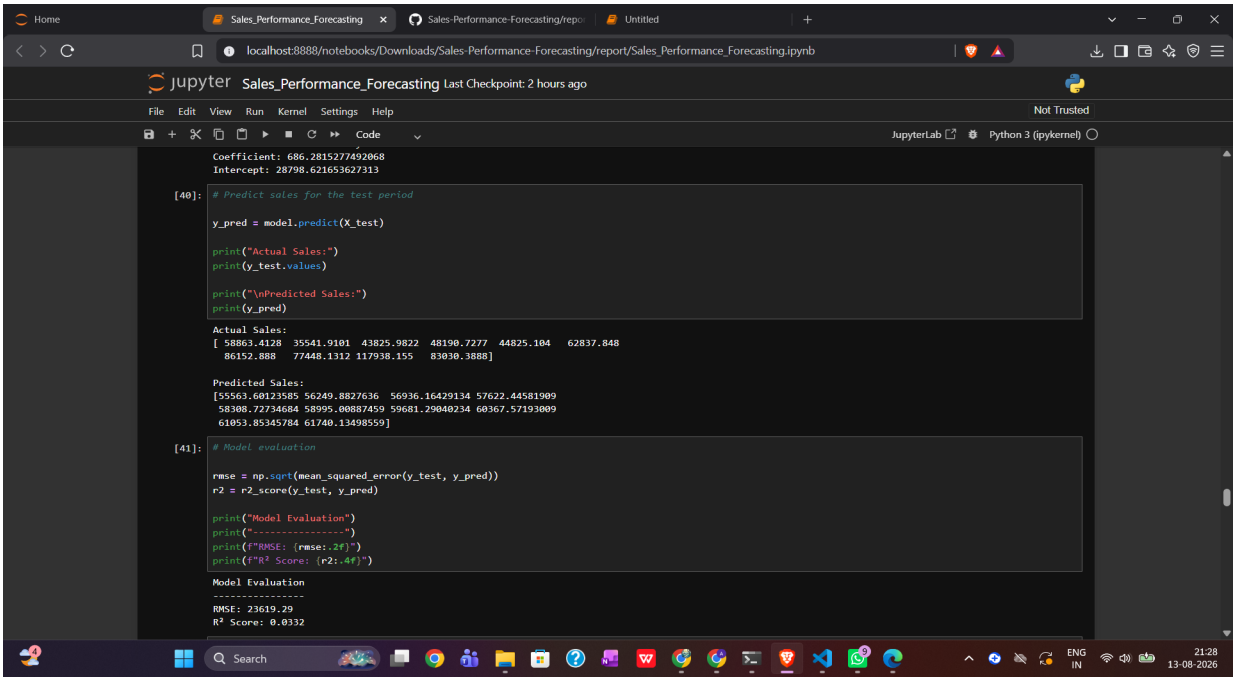

Figure 5 — Linear Regression Output

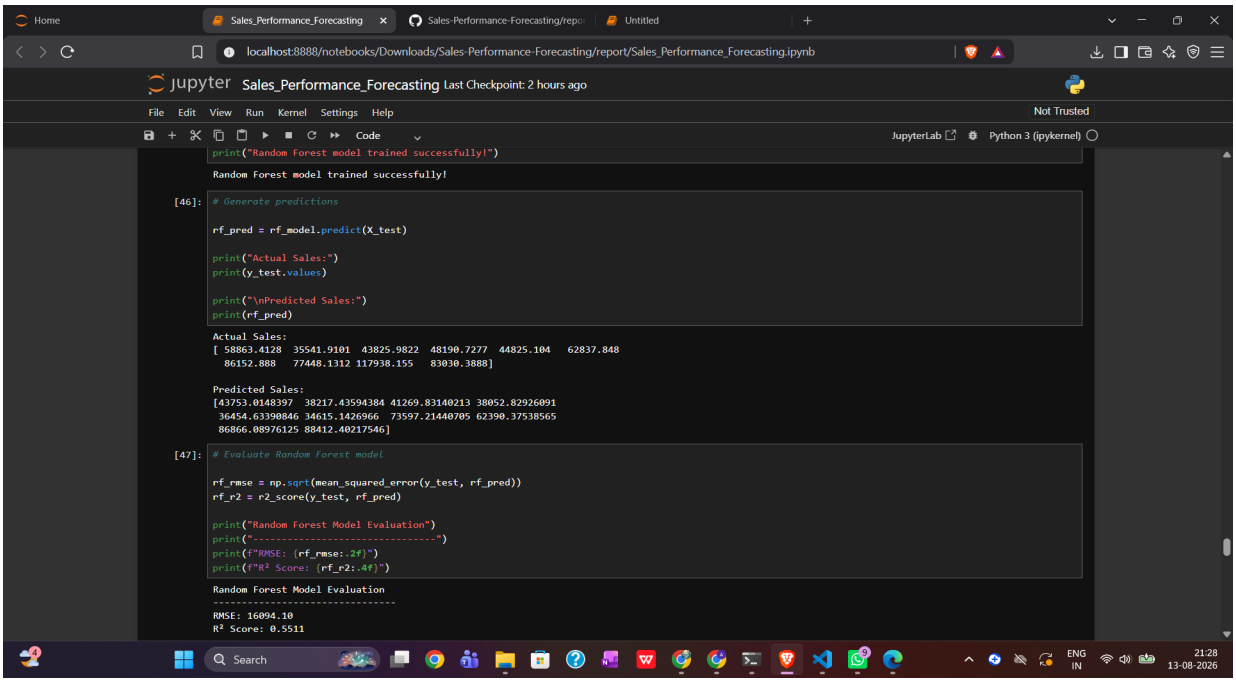

Figure 6 — Random Forest Output

4. Exploratory Data Analysis

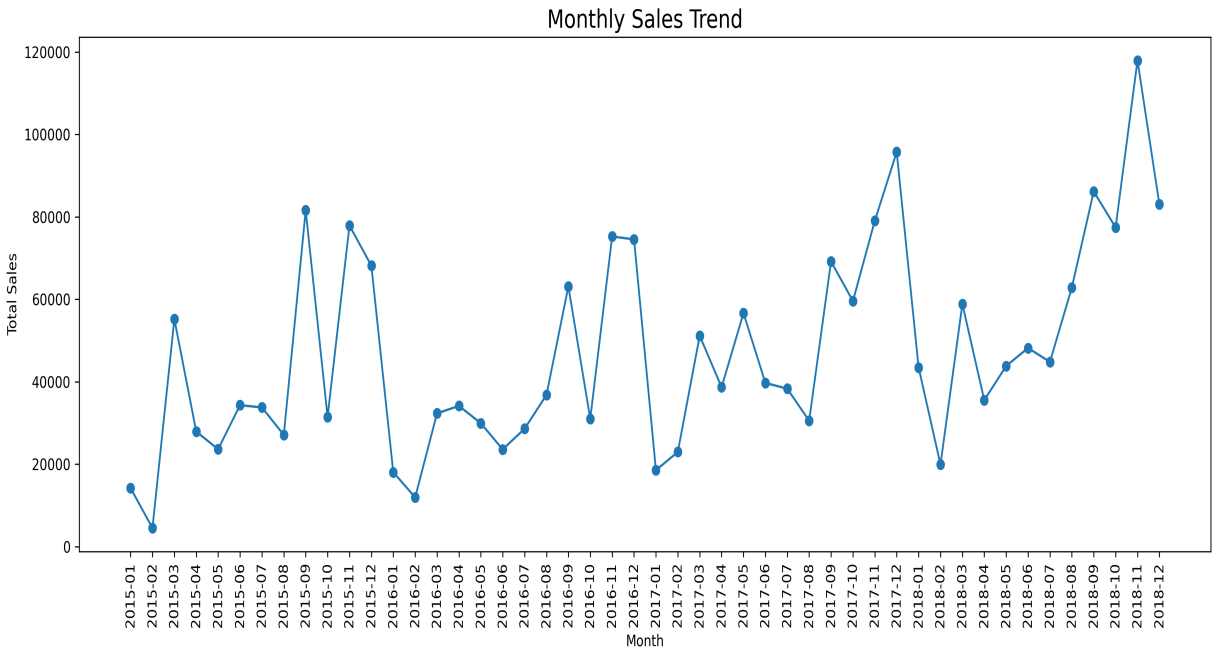

Figure 1 — Monthly Sales Trend

Interpretation: The monthly sales trend shows that sales change quite a lot from month to month. Some months have much higher sales than others, which shows that sales are not consistent throughout the year. November 2018 had the highest monthly sales, while February 2015 had the lowest.

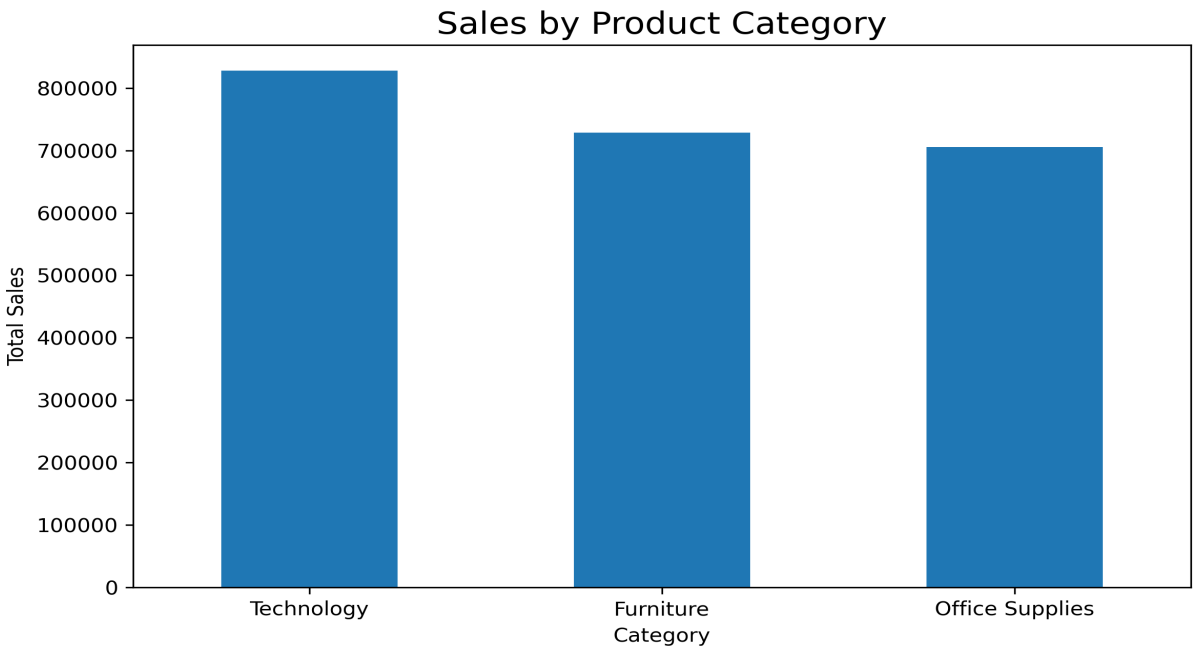

Figure 2 — Sales by Product Category

Interpretation: Technology is the highest-performing category with total sales of 827,455.87. Furniture and Office Supplies also contribute a large amount to total sales. This shows that Technology products are an important part of the business.

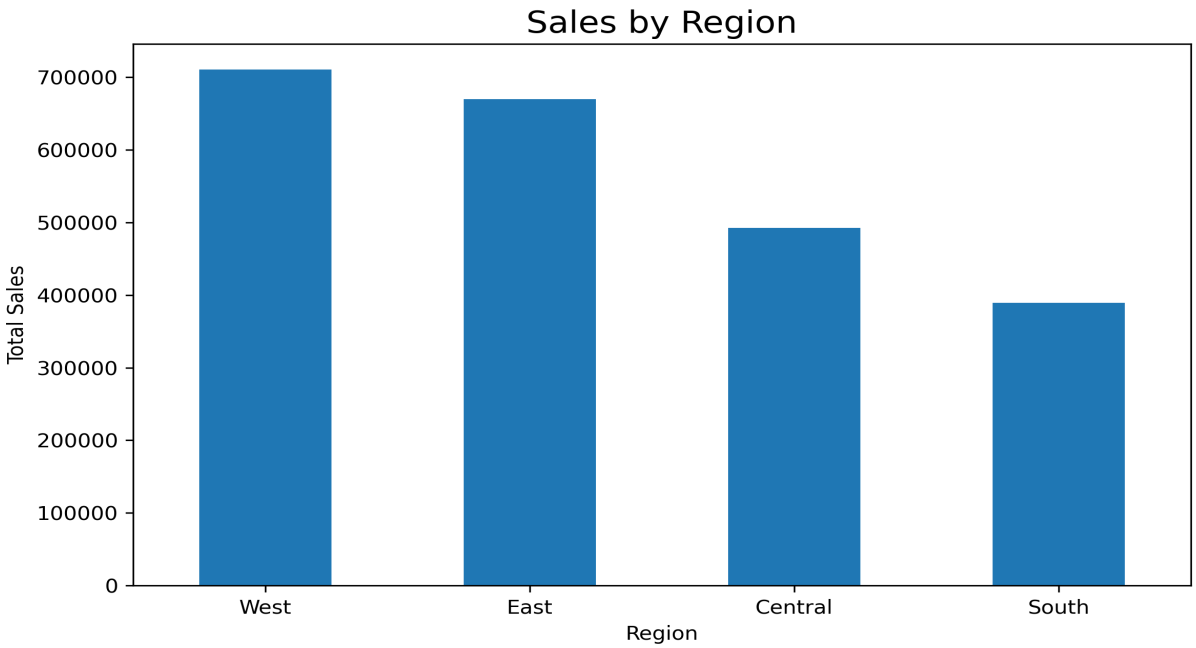

Figure 3 — Sales by Region

Interpretation: The West region has the highest total sales at 710,219.68. The performance of the regions is different, with some regions contributing much less than the West. The lower-performing regions could be studied further to understand the reasons for the difference.

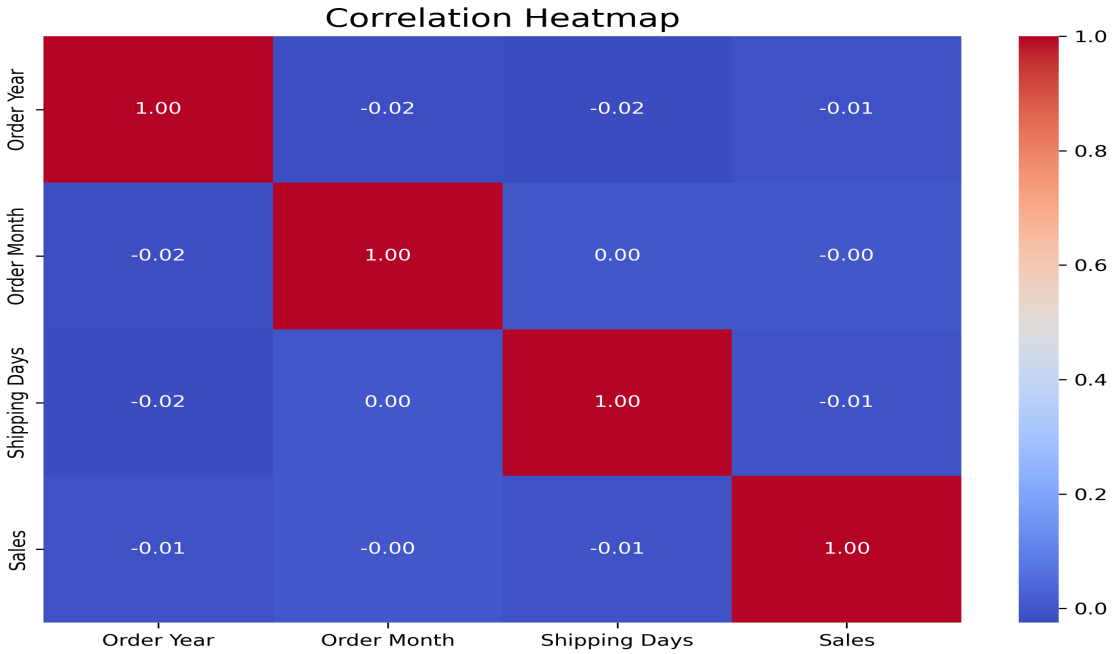

Figure 4 — Correlation Heatmap

Interpretation: The heatmap shows the relationships between the numerical variables in the dataset. It helps identify which variables have stronger or weaker relationships with Sales and Profit. These relationships can also help in deciding which variables may be useful for analysis and modelling.

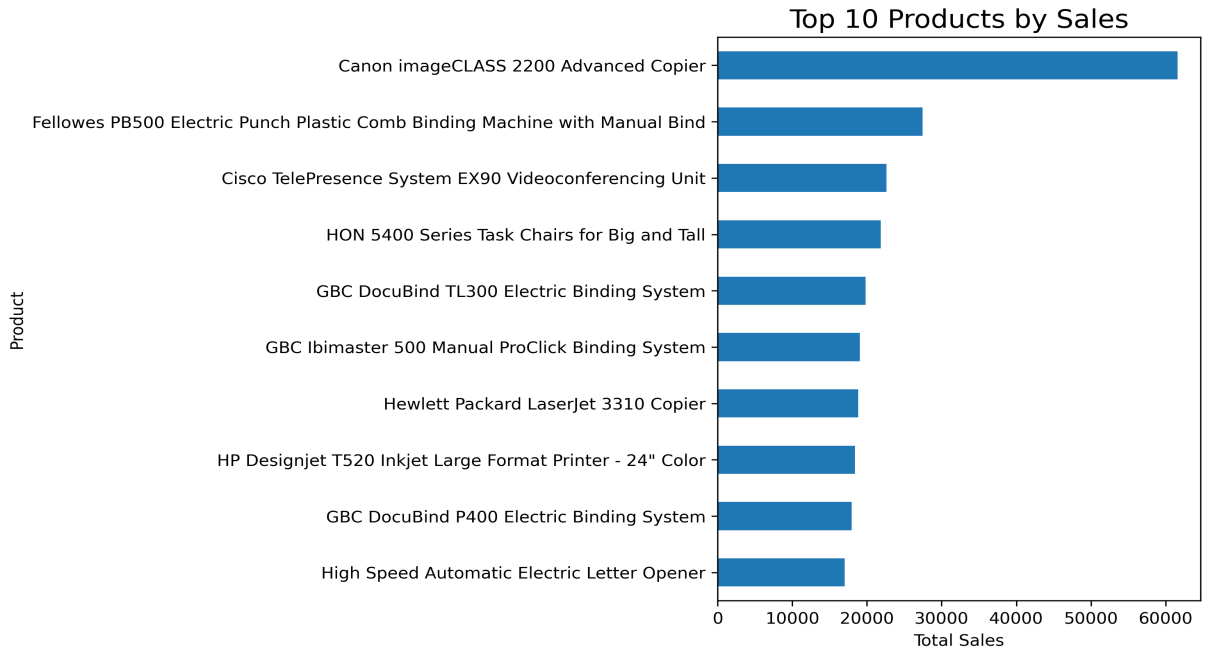

Figure 5 — Top 10 Products by Sales

Interpretation: The top 10 products contribute a significant amount of sales compared with other individual products. The Canon imageCLASS 2200 Advanced Copier is the top-selling product with total sales of 61,599.82. These high-performing products can be given more attention during inventory and sales planning.

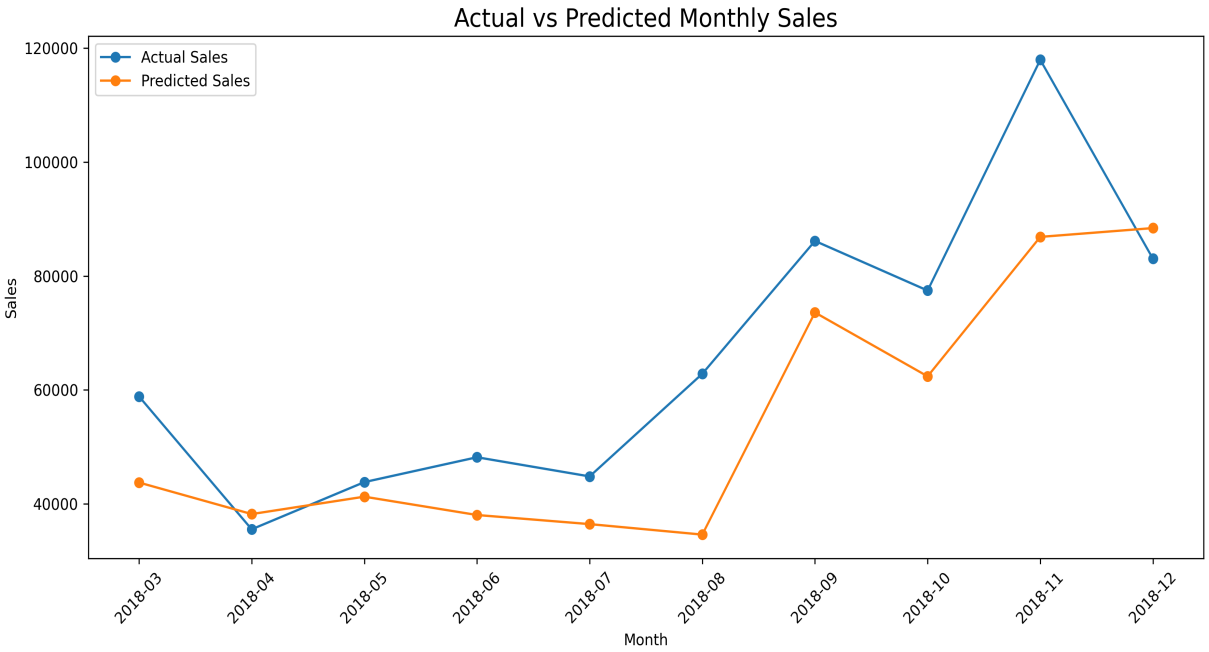

Figure 6 — Actual vs Predicted Monthly Sales

Interpretation: The chart compares the actual monthly sales with the values predicted by the model. The Random Forest predictions follow the general pattern of the actual sales better than the simple Linear Regression model. However, there are still differences between actual and predicted values, so the model can be improved further.

5. Key Insights

• **Technology performed the best:** Technology had the highest total sales of 827,455.87, followed by Furniture at 728,658.58 and Office Supplies at 705,422.33. This shows that Technology products contribute the most to overall sales.

• **Regional performance is different:** The West region had the highest sales at 710,219.68, while the South region had the lowest sales at 389,151.46. This difference shows that there is scope to improve sales performance in the lower-performing regions.

• **Sales change a lot over time:** Monthly sales ranged from 4,519.89 in February 2015 to 117,938.15 in November 2018. This shows that some months have much higher sales than others.

• **Top-selling product:** The Canon imageCLASS 2200 Advanced Copier was the highest-selling product, generating total sales of 61,599.82. High-performing products like this can be given more attention during inventory and sales planning.

6. Predictive Modeling

A Linear Regression model was first developed using monthly time sequence information. The model achieved an RMSE of 23,619.29 and an R^2 score of 0.0332. This shows that a simple linear trend was not able to capture the changes in monthly sales very well.

A Random Forest Regression model was then developed using time-based and seasonal features such as month number, year, month, and cyclical month features. The model achieved an RMSE of **16,094.10** and an R^2 score of **0.5511**.

Model	RMSE	R^2 Score
Linear Regression	23,619.29	0.0332
Random Forest	16,094.10	0.5511

Model Interpretation: Random Forest performed better than Linear Regression because it had a lower RMSE and a higher R^2 score. The R^2 value of 0.5511 means that the model was able to explain around 55% of the variation in the monthly sales data. However, there is still room for improvement in the forecasting results.

7. Business Recommendations

1. Focus more on Technology products for inventory and promotions because this category had the highest total sales.

2. Look into the lower sales of the South region and try to understand whether factors such as customer demand, product availability, pricing, or marketing are affecting its performance.

3. Give more attention to high-performing products such as the Canon imageCLASS 2200 Advanced Copier when planning inventory and promotions.

4. Use the monthly sales patterns to plan inventory, staffing, and marketing activities during periods when sales are expected to be higher.

5. The forecasting model can be improved in the future by adding more useful variables such as customer segment, discount, shipping mode, product category, and previous months' sales.

8. Conclusion

The Sales Performance & Forecasting Analysis identified important patterns in the Superstore dataset. Technology was the highest-performing category, while the West was the strongest region. Sales also showed substantial variation across months, with November 2018 recording the highest monthly sales.

The Random Forest model performed better than the initial Linear Regression model. The final Random Forest model achieved an R^2 score of 0.5511 and an RMSE of 16,094.10. The results show that adding time and seasonal information helped the model predict monthly sales better.

The model could be improved in the future by adding more useful features such as discounts, customer segments, product information, and previous months' sales.

9. GitHub Repository

The complete project files, Jupyter Notebook, dataset, charts, and final report are available on GitHub:

https://github.com/ankita7833/Sales-Performance-Forecasting/tree/main